VTPEH 6270 - Check Point 03

Data Set Selection & R markdown Report Outline

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1 Research Question and Background

1.1 Research Question

In 2023, how does the prevalence of arthritis among adults vary across U.S. states?

1.2 Scientific Plausibility and Context

Arthritis is one of the most common chronic conditions among U.S. adults and a major contributor to pain, disability, and reduced quality of life (Elgaddal et al. 2024).
Adults with arthritis are generally less physically active than those without, and many report engaging in only low-impact activities such as walking, which highlights the functional impact of the condition (Centers for Disease Control and Prevention 2021).
Counties with higher arthritis prevalence and arthritis-attributable pain outcomes show significant spatial clustering in the United States, indicating that geographic variation is not random (Sun, Zajacova, and Grol-Prokopczyk 2024).
Therefore, it is scientifically plausible that adult arthritis prevalence varies across U.S. states in 2023, reflecting differences in the distribution of higher and lower burden counties within states.

2 Data Preparation

```r
# Import data

library(tidyverse)
library(janitor)
library(readr)

setwd("/Users/eve/Desktop/V270")
places <- read_csv("PLACES__Local_Data_for_Better_Health,_County_Data,_2025_release_20260127.csv") |>
  clean_names()

#checking column names
names(places)

#checking data_value_type
places |>
  distinct(data_value_type) |>
  arrange(data_value_type) |>
  print(n = 50)

# Subset to analytic dataset for the research question
arthritis_2023 <- places |>
  filter(
    year == 2023,
    measure == "Arthritis among adults",
    data_value_type == "Crude prevalence"
  ) |>
  filter(!is.na(data_value)) |>
  select(
    state_desc, location_name,
    year, measure, data_value_type,
    data_value, low_confidence_limit, high_confidence_limit
  )

# Quick checks
arthritis_2023 |>
  summarise(
    n_rows = n(),
    n_states = n_distinct(state_desc),
    min = min(data_value),
    max = max(data_value)
  )
```

2.1 Derived Variables

```r
arthritis_2023 <- arthritis_2023 |>
  mutate(`arthritis_prop` = data_value / 100)

summary(arthritis_2023$arthritis_prop)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1270  0.2710  0.3000  0.2976  0.3270  0.4580
```

The research question compares crude arthritis prevalence (`data_value`) across states (`state_desc`) directly. I converted the crude prevalence from percent to proportion to facilitate interpretation.

2.2 Categorization and Discretization

No discretization was needed because the grouping variable `state_desc` is categorical (U.S. state), which supports grouped summaries and visualization; the outcome `data_value` is continuous (county-level crude prevalence).

3 Grouped Summary Statistics

3.1 Stratified Analysis

3.1.1 Table 1

```r
library(knitr)

table_AB <- arthritis_2023 |>
  filter(!state_desc %in% c("United States")) |>
  group_by(state_desc) |>
  summarise(
    n_counties = n(),
    mean_pct = mean(data_value, na.rm = TRUE),
    sd_pct = sd(data_value, na.rm = TRUE),
    median_pct = median(data_value, na.rm = TRUE),
    iqr_pct = IQR(data_value, na.rm = TRUE),
    min_pct = min(data_value, na.rm = TRUE),
    max_pct = max(data_value, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(median_pct)) |>
  mutate(across(where(is.numeric), ~ round(.x, 1)))

kable(
  table_AB,
  caption = "Descriptive statistics of county-level crude arthritis prevalence (percent) stratified by U
  col.names = c("State", "N counties", "Mean", "SD", "Median", "IQR", "Min", "Max"),
  booktabs = TRUE
)
```

Table 1: Descriptive statistics of county-level crude arthritis prevalence (percent) stratified by U.S. state, 2023 (PLACES).

State	N counties	Mean	SD	Median	IQR	Min	Max
West Virginia	55	39.8	3.3	40.5	3.5	25.8	45.8
Tennessee	95	34.1	3.4	34.5	3.9	24.4	41.5

State	N counties	Mean	SD	Median	IQR	Min	Max
Maine	16	34.0	3.0	33.8	3.4	27.2	39.7
Alabama	67	33.1	3.1	33.4	3.8	25.0	39.2
Michigan	83	33.0	4.1	32.9	5.3	22.1	41.5
Missouri	115	32.2	3.5	32.4	4.4	20.8	42.6
Mississippi	82	31.7	2.9	32.2	4.0	22.1	37.6
Arkansas	75	31.9	3.4	32.1	4.1	21.7	37.7
Montana	56	31.6	3.4	32.0	4.3	21.5	40.4
Ohio	88	31.8	2.7	32.0	3.4	24.5	37.8
Louisiana	64	31.6	3.1	31.6	4.4	25.5	40.4
Oregon	36	31.7	4.4	31.6	6.3	22.3	40.5
South Carolina	46	30.8	3.1	31.6	3.7	24.7	39.2
Oklahoma	77	30.6	2.9	31.4	3.7	21.4	37.2
North Carolina	100	30.6	3.7	31.3	4.5	19.7	37.9
Indiana	92	30.9	2.6	31.2	3.1	21.4	36.4
Virginia	133	30.3	4.9	30.9	6.2	18.8	39.9
North Dakota	53	30.3	3.7	30.7	3.7	21.3	36.4
Washington	39	29.3	5.3	30.0	7.4	18.4	39.2
Vermont	14	29.7	2.8	29.9	2.6	22.2	34.7
Wisconsin	72	29.6	3.4	29.8	5.2	19.5	35.5
Kansas	105	29.3	3.5	29.7	4.2	17.1	36.1
Nevada	17	30.2	4.1	29.7	6.5	23.9	37.1
Florida	67	29.4	4.2	29.6	4.0	20.6	41.5
Iowa	99	29.3	2.5	29.6	2.6	18.9	33.6
Illinois	102	29.2	3.0	29.5	3.3	20.8	35.4
Georgia	159	28.9	3.9	29.4	4.5	13.9	39.0
New Hampshire	10	29.4	2.9	29.1	4.5	25.8	33.8
Delaware	3	29.6	4.4	29.1	4.4	25.5	34.2
South Dakota	66	28.7	3.1	28.9	3.4	18.3	34.8
New York	62	27.9	3.2	28.7	4.5	20.3	34.6
Idaho	44	29.0	4.6	28.5	7.3	12.7	36.4
Nebraska	93	28.3	3.0	28.5	3.4	21.2	33.9
Arizona	15	28.3	5.3	28.1	8.5	21.8	39.0
Maryland	24	28.9	3.7	28.0	6.3	21.4	35.4
Colorado	64	27.5	4.0	27.6	5.9	18.3	35.3
New Mexico	33	28.1	4.4	27.5	5.7	21.7	38.8
Wyoming	23	27.3	3.4	27.3	4.3	18.6	33.2
Massachusetts	14	27.3	4.2	27.0	5.9	19.2	35.4
Rhode Island	5	26.9	2.0	26.9	1.8	23.7	28.7
Minnesota	87	25.8	2.9	26.6	3.6	18.7	33.1
Hawaii	5	27.0	4.8	26.5	2.8	21.8	34.8
Texas	253	26.3	4.0	26.5	5.9	17.6	38.6
Connecticut	9	25.6	2.7	26.3	2.6	21.5	29.3
Utah	29	26.2	4.1	26.2	5.5	17.1	35.1
Alaska	30	24.9	3.9	24.9	4.2	15.7	31.6
New Jersey	21	23.4	3.4	23.9	4.4	16.5	30.0
California	58	24.4	4.5	23.7	6.7	16.6	35.2
District of Columbia	1	17.4	NA	17.4	0.0	17.4	17.4

3.1.2 Interpertation

The stratified descriptive statistics show substantial variation in county-level crude arthritis prevalence across states. West Virginia has the highest median prevalence (40.5%), whereas the District of Columbia has the lowest (17.4%), indicating a large between-state difference. Several southeastern/Appalachian states appear among the highest medians, while states such as New Jersey and California are among the lowest. Overall, these results suggest that arthritis prevalence (A) is associated with state of residence (B).

4 Comparative Visualizations

4.1 Figure 1

Due to its size, Figure 1 is presented on page 6.

```r
library(dplyr)

# remove national summary row if present (keep states + DC)
plot_df <- arthritis_2023 |>
  dplyr::filter(!state_desc %in% c("United States"))

# order states by median prevalence for readability
state_order <- plot_df |>
  dplyr::group_by(state_desc) |>
  dplyr::summarise(med = median(data_value, na.rm = TRUE), .groups = "drop") |>
  dplyr::arrange(med) |>
  dplyr::pull(state_desc)

plot_df <- plot_df |>
  dplyr::mutate(state_desc = factor(state_desc, levels = state_order))

# boxplot
ggplot(plot_df, aes(x = state_desc, y = data_value)) +
  geom_boxplot(outlier.alpha = 0.2) +
  coord_flip() +
  labs(
    x = "State",
    y = "County-level crude arthritis prevalence (%)"
  )
```

4.1.1 Interpretation

Figure 1 indicates substantial geographic variation in adult arthritis prevalence across U.S. states in 2023. States such as West Virginia and Tennessee have markedly higher median county-level prevalence than states such as New Jersey and California, suggesting that the arthritis burden is not evenly distributed across the country. The differing box widths and whisker lengths show that within-state heterogeneity also varies, with some states exhibiting much larger county-to-county differences than others. Overall, the pattern supports the research question by demonstrating clear state-level differences in arthritis prevalence and highlights that both between-state and within-state factors may contribute to observed variation.

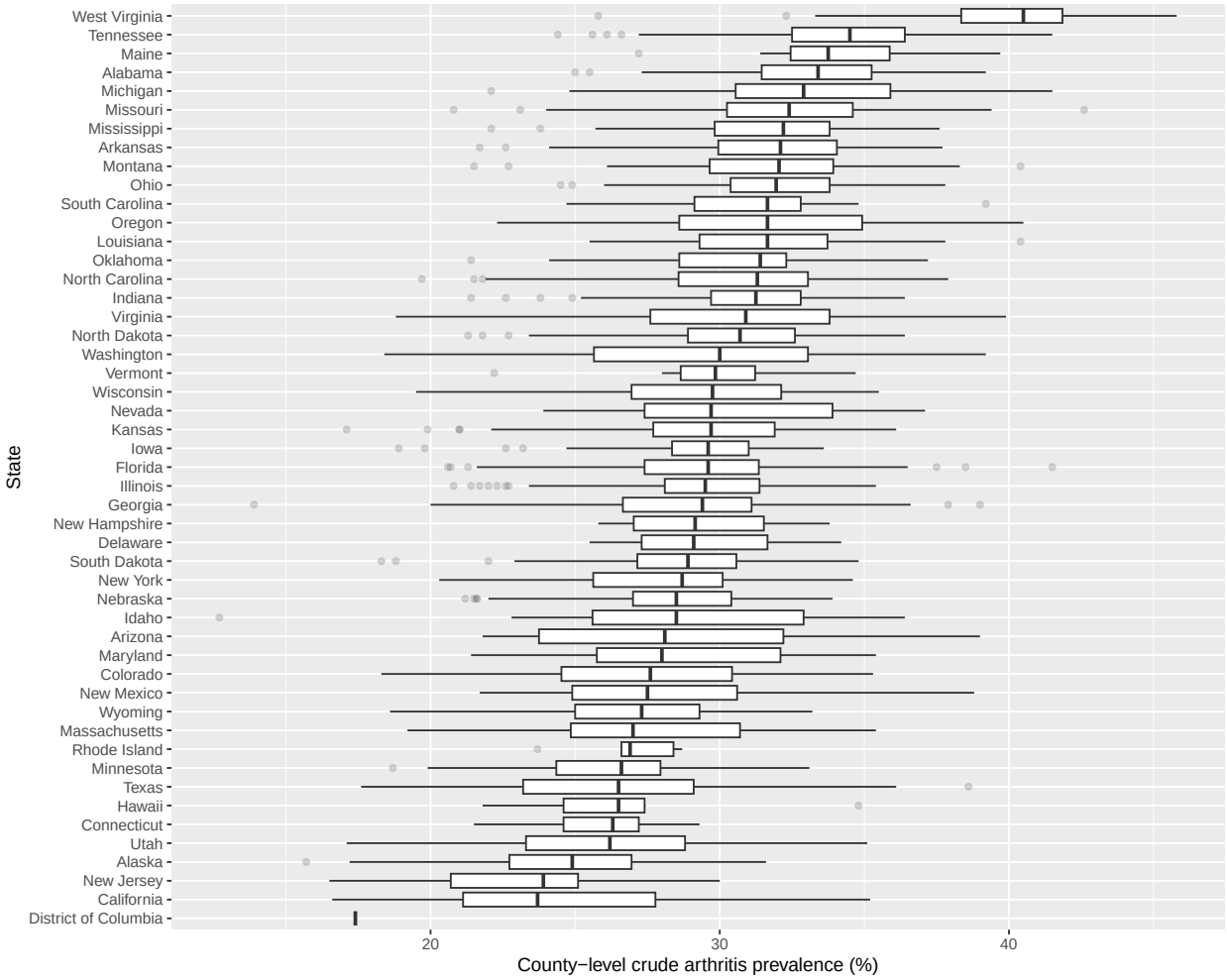

Figure 1: County-level crude arthritis prevalence by state in 2023. Boxes represent the interquartile range (IQR), the central line indicates the median, and whiskers extend to 1.5 × IQR. States are ordered by median prevalence. Data are from the CDC PLACES 2023 release.

5 AI Use Disclosure Statement

ChatGTP was used only when to generate R codes.

References

Centers for Disease Control and Prevention. 2021. "Walking and Other Physical Activities — Adults with Arthritis." CDC. https://www.cdc.gov/mmwr/volumes/70/wr/mm7040a3.htm.

Elgaddal, Nazik, Ellen A. Kramarow, Julie D. Weeks, and Cynthia Reuben. 2024. "Arthritis in Adults Age 18 and Older: United States, 2022." Data Brief, no. 497. National Center for Health Statistics. https://www.cdc.gov/nchs/data/databriefs/db497.pdf.

Sun, Feinuoa, Anna Zajacova, and Hanna Grol-Prokopczyk. 2024. "The Geography of Arthritis-Attributable Pain Outcomes: A County-Level Spatial Analysis." *PAIN* 165 (7): 1505–12. https://doi.org/10.1097/j.pain.0000000000003155.